

# Tables

Companion document to the manuscript “Verifiable, accountable, and reproducible AI-assisted research: the VAR framework and the AIVS verification schema, demonstrated on an end-to-end protein-variant workflow” (Ribbeck and Kiran, submitted to *Patterns*).

The three tables below are also embedded in the main manuscript (§3.3, §3.4, §3.5); they are reproduced here as a separate document for production-side typesetting.

## Table 1

**Table 1.** AIVS record types, workflow stage, key fields, and typical cardinality per manuscript.

| Record     | Workflow stage | Key fields                                                                                                          | Cardinality              |
|------------|----------------|---------------------------------------------------------------------------------------------------------------------|--------------------------|
| Source     | Source         | id, type, uri, content_hash, retrieved_at, license                                                                  | One or more per workflow |
| Invocation | Generate       | id, parent_id, tool, model, version, prompt, parameters, response, operator, access_date, retention_policy, purpose | One per AI call          |
| Evaluation | Evaluate       | id, invocation_ref, check_type, target, result, confidence, verifier, evidence_uri                                  | Many per Invocation      |
| Curation   | Curate         | id, invocation_refs, kept, rationale, accountable_author                                                            | One or more per workflow |

| Record     | Workflow stage | Key fields                                                                  | Cardinality                |
|------------|----------------|-----------------------------------------------------------------------------|----------------------------|
| Deposition | Deposit        | id, artifact_type, uri, repository, doi, version, license, retention_policy | One per published artifact |

## Table 2

**Table 2.** AIVS submission-time verification checks, the failure mode each addresses, and the reviewer judgment that remains.

| Check                   | Failure mode addressed               | Reviewer judgment retained                             |
|-------------------------|--------------------------------------|--------------------------------------------------------|
| Citation resolution     | Hallucinated references              | Whether cited work supports the claim                  |
| Quote matching          | Misattributed or invented quotations | Whether quoted material is interpreted accurately      |
| Data retrievability     | Phantom datasets                     | Whether data is sufficient for the claim               |
| Code execution          | Non-runnable analysis                | Whether code implements the described method correctly |
| Image forensic analysis | Fabricated or manipulated figures    | Whether figures support the textual claims             |
| Schema validation       | Incomplete provenance                | Whether the documented workflow is appropriate         |

## Table 3

**Table 3.** AIVS disclosure tiers, access scope, and floor fields.

| Tier   | Public access       | Editorial/reviewer access | Floor fields (always public)                |
|--------|---------------------|---------------------------|---------------------------------------------|
| Public | Full record content | Full record content       | Model ID, version, parameters, verification |

| Tier      | Public access                | Editorial/reviewer access                | Floor fields (always public) |
|-----------|------------------------------|------------------------------------------|------------------------------|
|           |                              |                                          | results, accountable_author  |
| Editorial | Existence and rationale only | Full record content                      | As above                     |
| Redacted  | Redaction methodology only   | Redacted content under controlled access | As above                     |